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(54) **VEHICLE-MOUNTED AND HOUSEHOLD DUAL-PURPOSE MAGNETIC, ROTATABLE AND FOLDABLE MOBILE PHONE HOLDER CAPABLE OF BEING FREELY ADSORBED**

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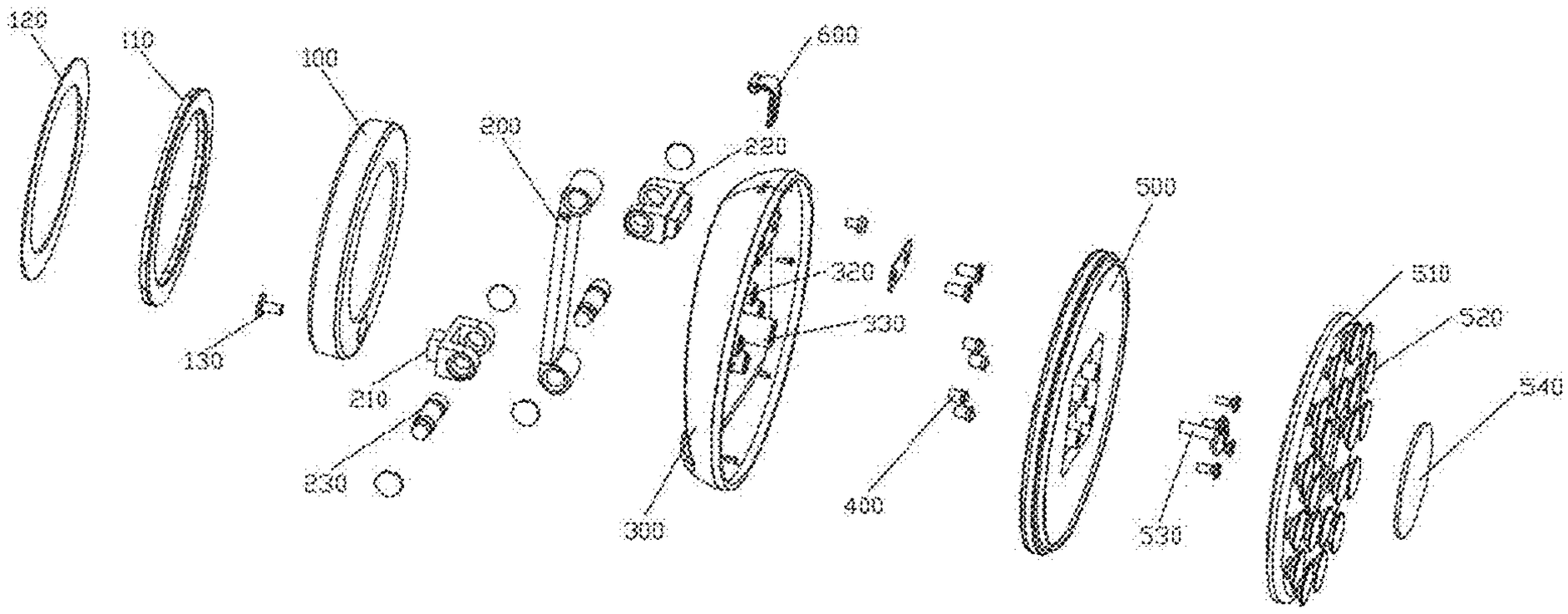
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CPC ***H04M 1/04*** (2013.01); ***F16M 11/041*** (2013.01); ***F16M 11/125*** (2013.01); ***F16M 13/022*** (2013.01)

(57) **ABSTRACT**

The present disclosure relates to a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed. The mobile phone holder includes a ring body, a housing and a bottom housing. A connection position is arranged in a bottom of the housing, a silica gel sucker is arranged at a bottom of the bottom housing, and the bottom housing and the silica gel sucker are connected to the connection position through pins. The ring body is provided with a magnet, the magnet is provided with a panel, and the ring body is connected to a swing rod. Two ends of the swing rod are connected to a first connecting block or a second connecting block through a rotating shaft, the first connecting block is connected to the ring body through a screw, and the second connecting block is connected to the housing through screws.



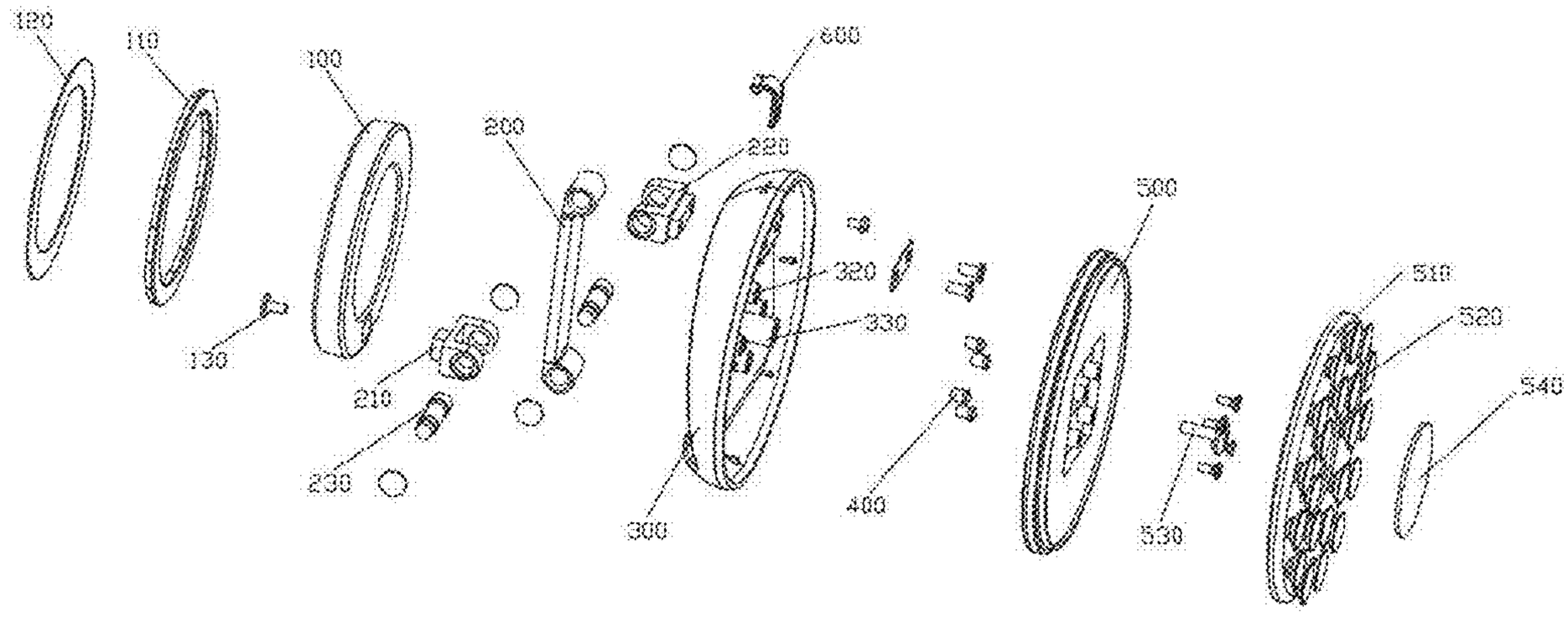

120
110
100
130
210
230
200
600
220
320
330
300
400
500
510
520
530
540

FIG. 1

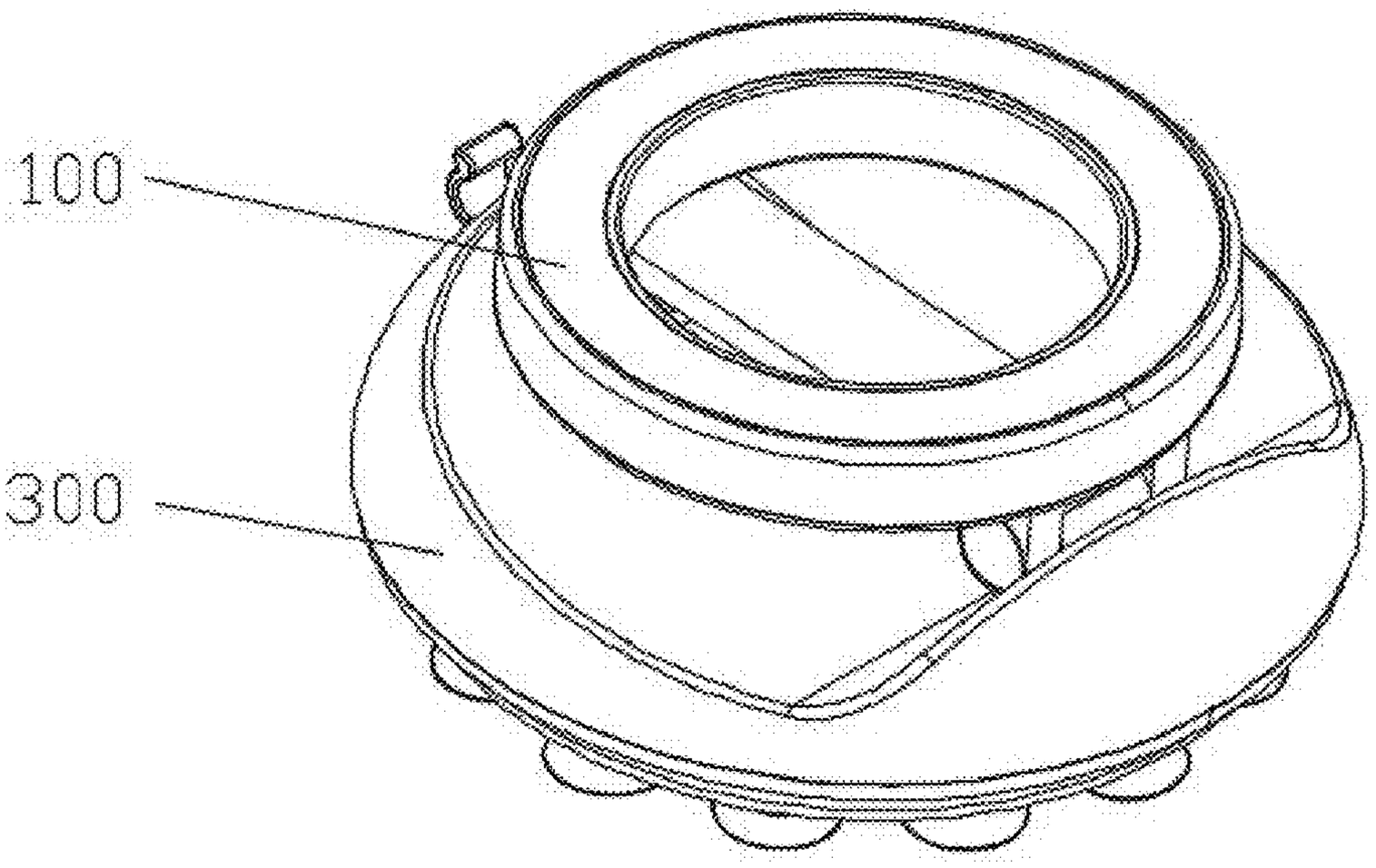

100
300

FIG. 2

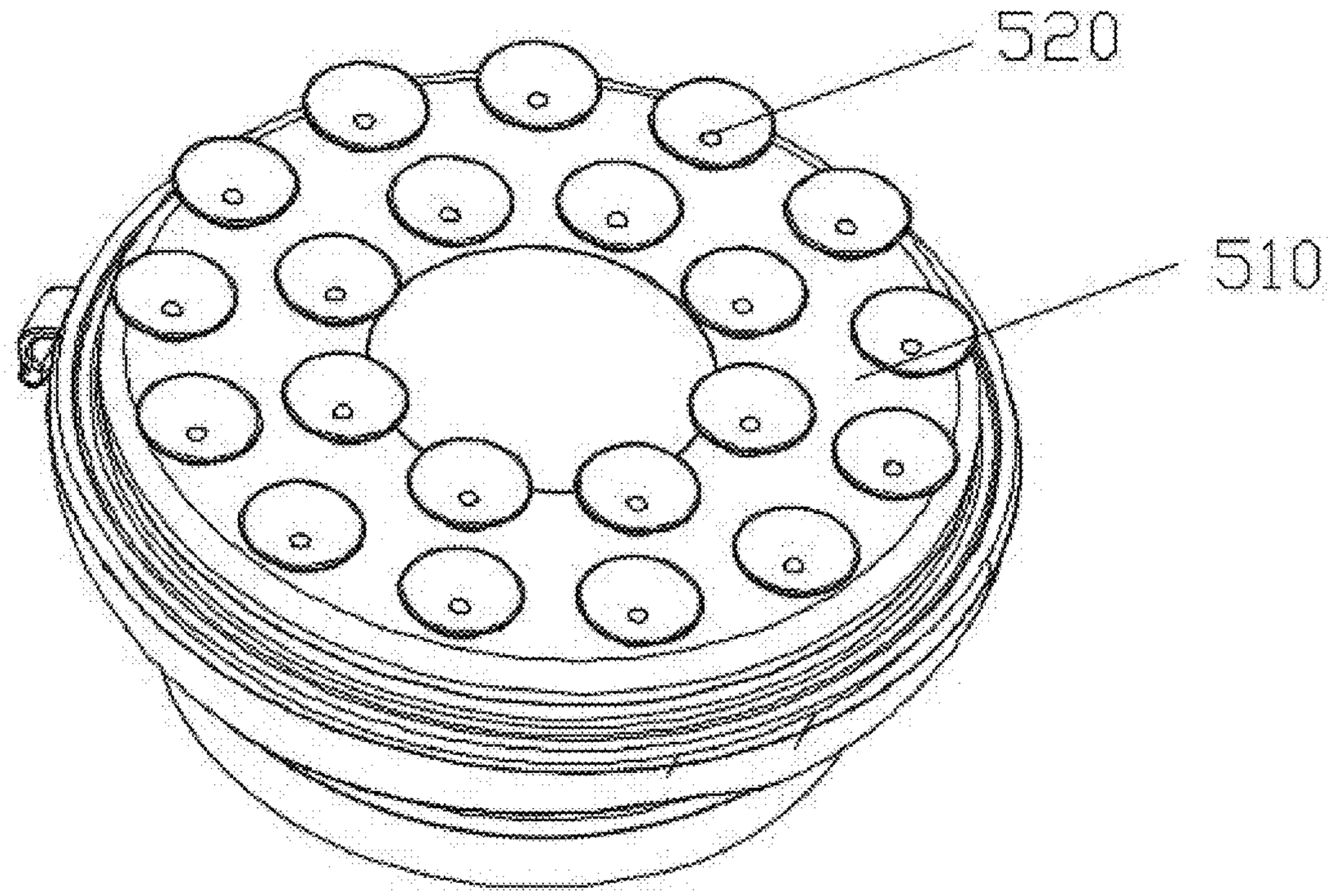


FIG. 3

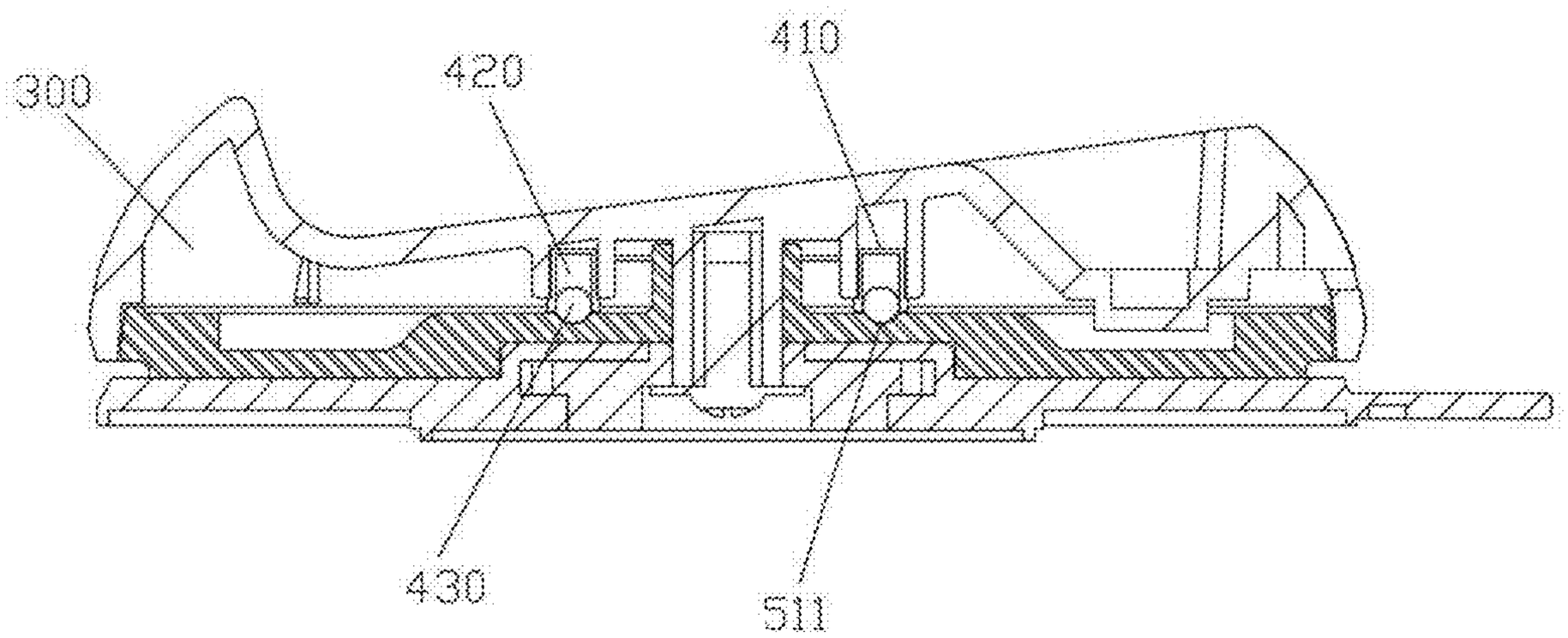


FIG. 4

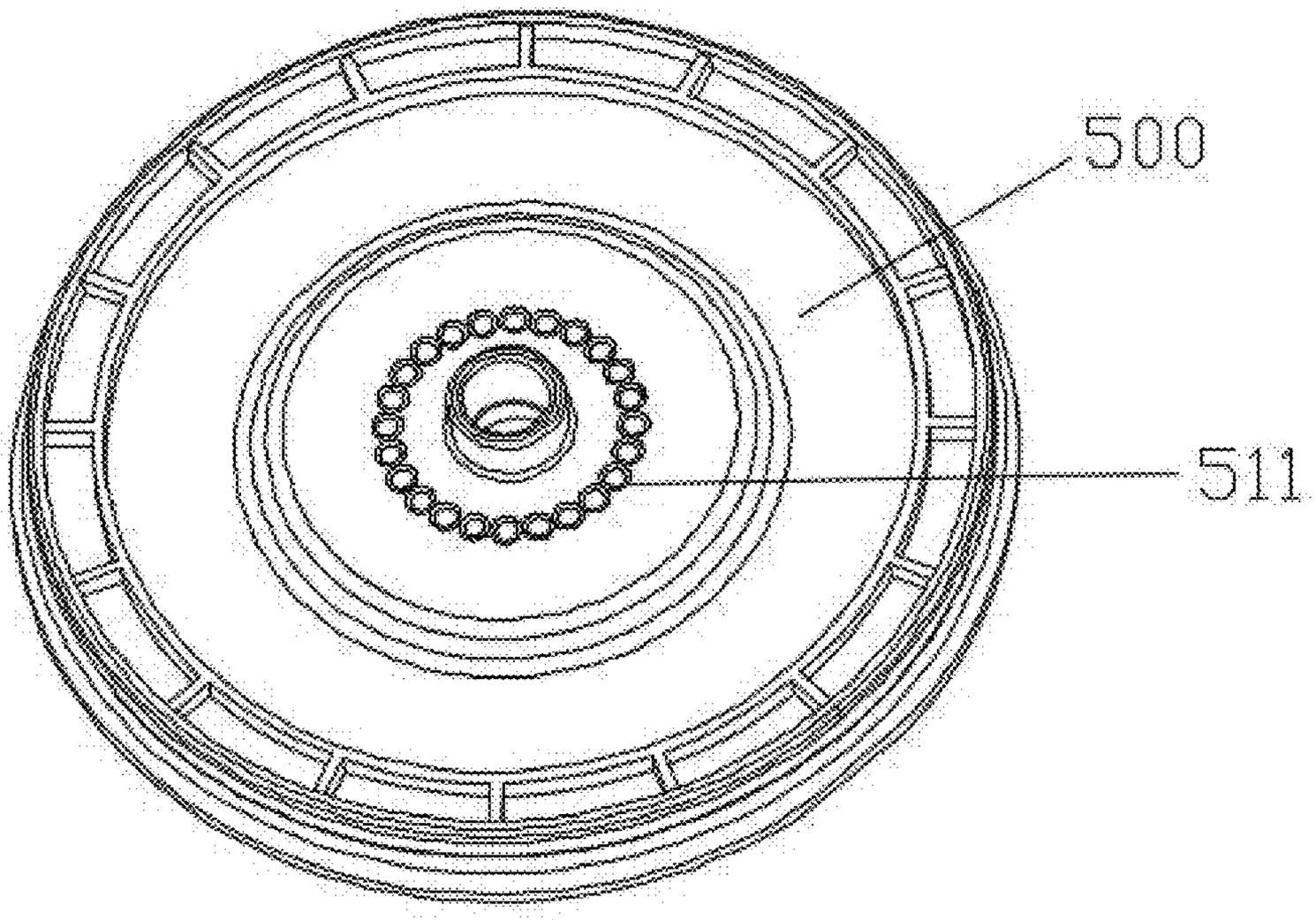


FIG. 5

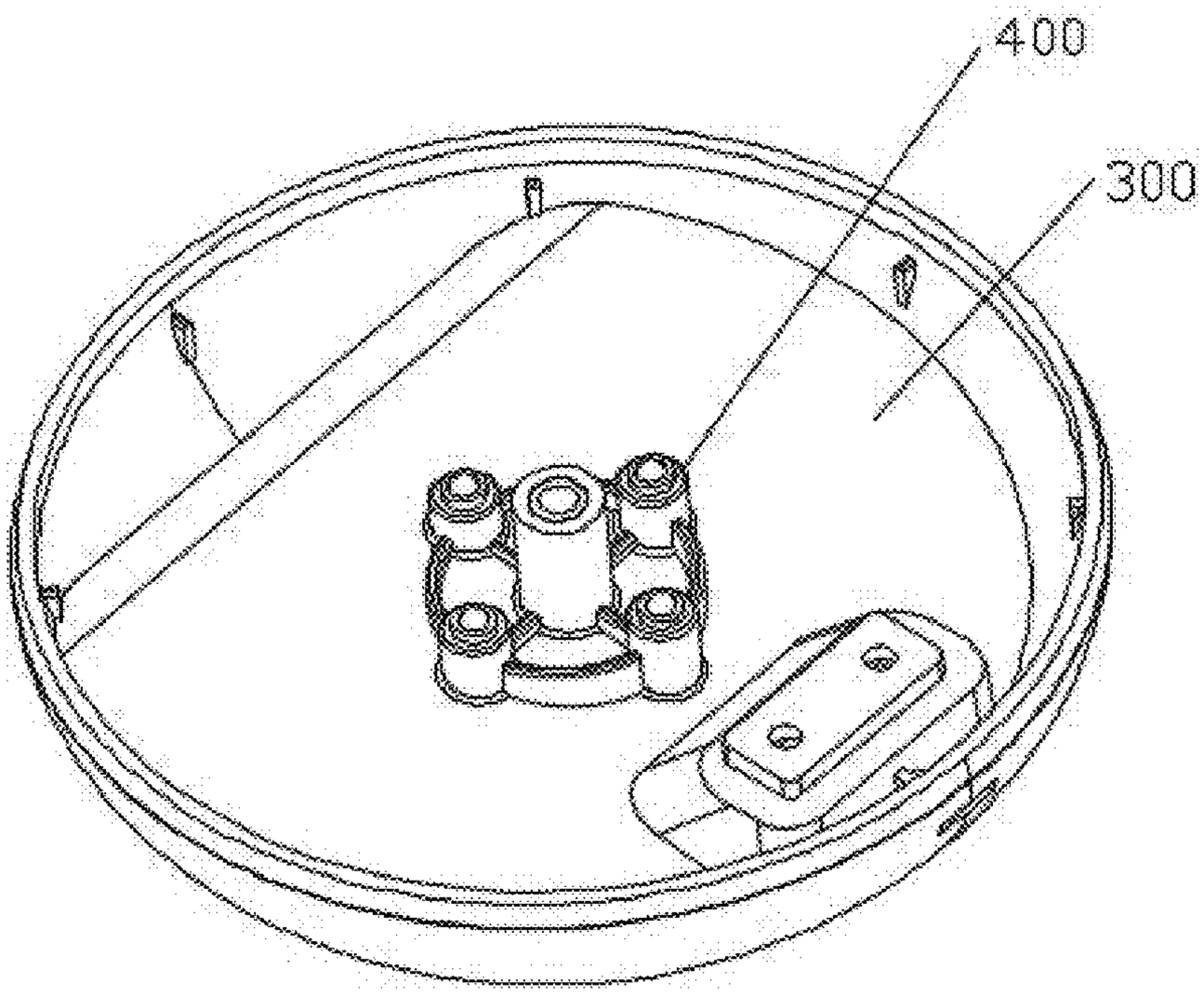


FIG. 6

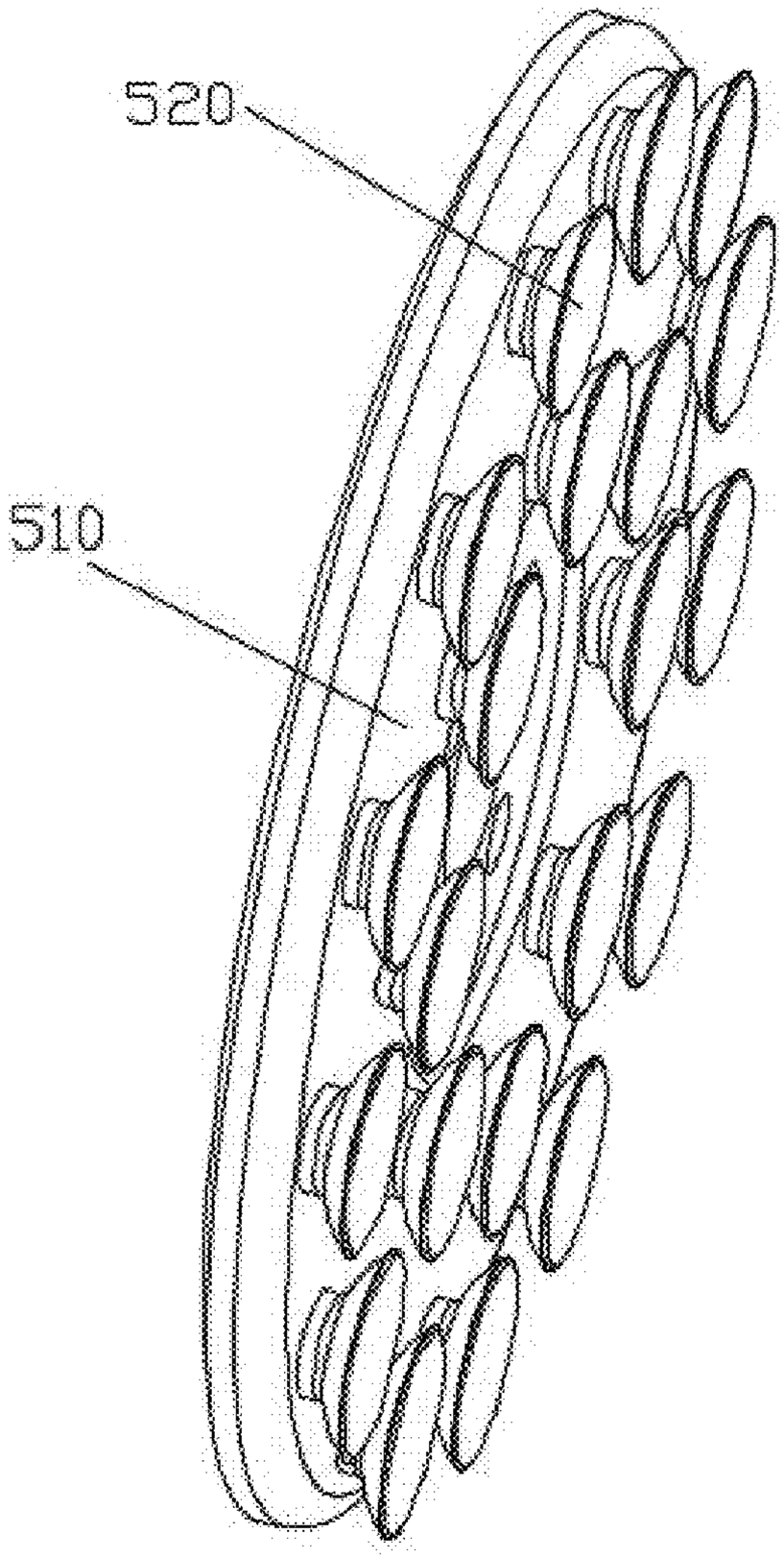


FIG. 7

VEHICLE-MOUNTED AND HOUSEHOLD DUAL-PURPOSE MAGNETIC, ROTATABLE AND FOLDABLE MOBILE PHONE HOLDER CAPABLE OF BEING FREELY ADSORBED

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to Chinese Patent Application No. 202421551172.3, filed on Jul. 3, 2024, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of vehicle-mounted furniture, and in particular to a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed.

BACKGROUND

[0003] At present, magsafe magnetic holders in a use environment of vehicle-mounted holders capable of being stuck on the market are inconvenient for angle adjustment and incompatible with POPsockets (airbag ring holder) and Ring (ordinary ring holder). Generally, holders capable of being stuck are only stuck on one type of products, which is inapplicable to products that require different use scenarios.

[0004] A vehicle-mounted holder is usually fixed on a plane console of a vehicle with glue. When a use position needs to be changed, dismounting is inconvenient due to glue, and it is difficult to clean glue residue.

[0005] Therefore, a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being stuck is invented.

SUMMARY

[0006] An objective of the present disclosure is to provide a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed, so as to solve the problems mentioned in the background art.

[0007] To achieve the above objective, the present disclosure provides the technical solutions as follows:

[0008] A vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed, including a ring body, a housing and a bottom housing, where a connection position is arranged in a center of a bottom of the housing, a silica gel sucker is arranged at a bottom of the bottom housing, and the bottom housing and the silica gel sucker are connected to the connection position through pins.

[0009] The ring body is fixedly provided with a magnet, the magnet is provided with a panel, and the ring body is connected to a swing rod.

[0010] Two ends of the swing rod are connected to a first connecting block or a second connecting block through a rotating shaft, the first connecting block is connected to the ring body through a screw, and the second connecting block is connected to the housing through screws.

[0011] An end, farthest away from the bottom housing, of the silica gel sucker is fixedly provided with suction cups, and a decorative plate is arranged in a middle of the silica gel sucker.

[0012] In a preferred embodiment of the present disclosure, insertion holes are formed at a circumference of the connection position, and a ball plunger is mounted inside each insertion hole.

[0013] In a preferred embodiment of the present disclosure, each ball plunger is composed of a ball retainer housing, a spring and a ball, the ball retainer housing is fixedly mounted in the insertion hole, the spring is mounted inside the ball retainer housing, the spring is compressed and connected to the ball, and the ball is clamped on the ball retainer housing.

[0014] An inner wall of the bottom housing is circumferentially provided with positioning grooves, and the balls extend out of the ball retainer housings and are connected to the positioning grooves.

[0015] In a preferred embodiment of the present disclosure, a cable clip is fixedly mounted at a top of the housing.

[0016] Additional aspects and advantages of the present disclosure will be set forth partially in the following description, which will become obvious in the following description, or may be learned by practice of the present disclosure.

[0017] The ring body is fixedly provided with the magnet, the magnet is provided with the panel, and the ring body is connected to the swing rod. The ring body at an upper portion can be compatible with POPsockets (airbag ring holder) and Ring (ordinary ring holder). The hollow design of the ring body is compatible with POPsockets (airbag ring holder) and Ring (ordinary ring holder) for use and is applicable to bare mobile phones of magsafe Iphone mobile phones and magsafe magnetic housings, and the mobile phone holder can be used in combination with ring iron sheets for mobile phones of other models.

[0018] The housing and the bottom housing are tightly connected through the connection position and the pins, such that rotation of the mobile phone holder is realized. The balls are designed inside the housing and are in a compressed state. The balls are connected into the positioning grooves to realize the effect of keeping an angle unchanged after rotation, that is, the bottom housing can realize a high-angle rotation and hovering function.

[0019] The end, farthest away from the bottom housing, of the silica gel sucker is fixedly provided with the suction cups, and the suction cups on the silica gel sucker can be directly adsorbed on a plane workbench of a vehicle.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] The above-mentioned and/or additional aspects and advantages of the present disclosure will become apparent and easy to understand from the description of the examples with reference to the following accompanying drawings, in which:

[0021] FIG. **1** is a schematic exploded diagram of a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed.

[0022] FIG. **2** is a first perspective diagram of a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed.

[0023] FIG. **3** is a second perspective diagram of a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed.

[0024] FIG. **4** is sectional view of a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed.

[0025] FIG. **5** is a schematic structural diagram of a bottom housing in a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed.

[0026] FIG. **6** is a schematic structural diagram of a ball plunger in a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed.

[0027] FIG. **7** is a perspective diagram of a silica gel sucker in a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed.

[0028] In the figures: **100**. ring body; **110**. magnet; **120**. panel. **130**. screw; **200**. swing rod; **210**. first connecting block; **220**. second connecting block; **230**. rotating shaft; **300**. housing; **320**. insertion hole; **330**. connection position; **400**. ball plunger; **410**. ball retainer housing; **420**. spring; **430**. ball; **500**. bottom housing; **510**. silica gel sucker; **511**. positioning groove; **520**. suction cup; **530**. pin; **540**. decorative plate; and **600**. cable clip.

DETAILED DESCRIPTIONS OF THE EMBODIMENTS

[0029] Examples of the present disclosure are described in detail below, instances of the examples are shown in the accompanying drawings, throughout which identical or similar reference numerals denote identical or similar elements or elements having identical or similar functions. The examples described below with reference to the accompanying drawings are exemplary and intended to explain the present disclosure, instead of being construed as limiting the present disclosure.

[0030] Referring to FIGS. **1-7**, a vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed, including a ring body **100**, a housing **300** and a bottom housing **500**, where a connection position **330** is arranged in a center of a bottom of the housing **300**, a silica gel sucker **510** is arranged at a bottom of the bottom housing **500**, and the bottom housing **500** and the silica gel sucker **510** are connected to the connection position **330** through pins **530**.

[0031] The ring body **100** is fixedly provided with a magnet **110**, the magnet **110** is provided with a panel **120**, and the ring body **100** is connected to a swing rod **200**. The swing rod **200** can realize an up-and-down folding function, and the ring body **100** at an upper portion can be compatible with POPsockets (airbag ring holder) and Ring (ordinary ring holder). The hollow design of the ring body **100** is compatible with POPsockets (airbag ring holder) and Ring (ordinary ring holder) for use and is applicable to bare mobile phones of magsafe Iphone mobile phones and magsafe magnetic housings, and the mobile phone holder can be used in combination with ring iron sheets for mobile phones of other models.

[0032] Two ends of the swing rod **200** are connected to a first connecting block **210** or a second connecting block **220** through a rotating shaft **230**, the first connecting block **210** is connected to the ring body **100** through a screw **130**, and the second connecting block **220** is connected to the housing **300** through screws. The rotating shaft **230** is arranged as a damping rotating shaft and has a 360-degree rotation and hovering function, such that the swing rod can swing, and a component at the upper portion can hover.

[0033] Insertion holes **320** are formed at a circumference of the connection position **330**, and a ball plunger **400** is mounted inside each insertion hole **320**. Each ball plunger **400** includes a ball retainer housing **410**, a spring **420** and a ball **430**, the ball retainer housing **410** is fixedly mounted in the insertion hole **320**, the spring **420** is mounted inside the ball retainer housing **410**, the spring **420** is compressed and connected to the ball **430**, and the ball **430** is clamped on the ball retainer housing **410**.

[0034] An inner wall of the bottom housing **500** is circumferentially provided with positioning grooves **511**, and the balls **430** extend out of the ball retainer housings **410** and are connected to the positioning grooves **511**, thereby realizing a rotation function of the mobile phone holder. The housing **300** and the bottom housing **500** are tightly connected through the connection position **330** and the pins **530**, such that rotation of the mobile phone holder is realized. The balls **430** are designed inside the housing **300** and are in a compressed state. The balls **430** are connected into the positioning grooves **511** to realize the effect of keeping an angle unchanged after rotation, that is, the bottom housing **500** can realize a 360-degree rotation and hovering function.

[0035] A cable clip **600** is fixedly mounted at a top of the housing **300**.

[0036] An end, farthest away from the bottom housing **500**, of the silica gel sucker **510** is fixedly provided with suction cups **520**, and a decorative plate **540** is arranged in a middle of the silica gel sucker **510**.

[0037] The suction cups **520** on the silica gel sucker **510** can be directly adsorbed on a plane workbench of a vehicle.

[0038] The working principle of the present disclosure is: POPsockets (airbag ring holder), Ring (ordinary ring holder) or a mobile phone housing with metal are adsorbed on the ring body **100**, and the swing rod **200** can realize the up-and-down folding function. The balls **430** are connected into the positioning grooves **511** by rotating the housing **300** or the bottom housing **500**, that is, the bottom housing **500** can realize the 360-degree rotation and hovering function. A cable of the mobile phone can be fixed on the cable clip **600**. The end, farthest away from the bottom housing **500**, of the silica gel sucker **510** is fixedly provided with the suction cups **520**, and the suction cups **520** on the silica gel sucker **510** can be directly adsorbed on the plane workbench of the vehicle.

[0039] Although the examples of the present disclosure have been illustrated and described, it should be understood that those of ordinary skill in the art may make various changes, modifications, replacements and variations to these examples without departing from the principle and purpose of the present disclosure, and the scope of the present disclosure is limited by the appended claims and their legal equivalents.

What is claimed is:

1. A vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed, comprising a ring body, a housing and a bottom housing, wherein a connection position is arranged in a center of a bottom of the housing, a silica gel sucker is arranged at a bottom of the bottom housing, and the bottom housing and the silica gel sucker are connected to the connection position through pins;

the ring body is fixedly provided with a magnet, the magnet is provided with a panel, and the ring body is connected to a swing rod;

two ends of the swing rod are connected to a first connecting block or a second connecting block through a rotating shaft, the first connecting block is connected to the ring body through a screw, and the second connecting block is connected to the housing through screws; and

an end, farthest away from the bottom housing, of the silica gel sucker is fixedly provided with suction cups, and a decorative plate is arranged in a middle of the silica gel sucker.

2. The vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed according to claim **1**, wherein insertion holes are formed at a circumference of the connection position, and a ball plunger is mounted inside each insertion hole.

3. The vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed according to claim **2**, wherein each ball plunger comprises a ball retainer housing, a spring and a ball, the ball retainer housing is fixedly mounted in the insertion hole, the spring is mounted inside the ball retainer housing, the spring is compressed and connected to the ball, and the ball is clamped on the ball retainer housing; and

an inner wall of the bottom housing is circumferentially provided with positioning grooves, and the balls extend out of the ball retainer housings and are connected to the positioning grooves.

4. The vehicle-mounted and household dual-purpose magnetic, rotatable and foldable mobile phone holder capable of being freely adsorbed according to claim **1**, wherein a cable clip is fixedly mounted at a top of the housing.

* * * * *